Acanthamoeba Keratitis (MONDO:0005629): A Comprehensive Disease Characteristics Report

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> **Causal gene — stated up front:** Acanthamoeba keratitis (AK) is **not a genetic/Mendelian disease and has no human causal gene.** It is an **infectious disease** caused by free-living protozoa of the genus ***Acanthamoeba*** (phylum Amoebozoa; predominant genotype **T4**, ~66% of isolates). The template's "Causal gene(s): not recorded / OMIM: not recorded" is therefore correct and expected. Where this report cites "genes," they are **host immune-susceptibility *modifier* genes** (e.g., *CXCL8/IL-8*, *TLR4*, *IL17A*, *CLEC4E*) or **parasite virulence genes/proteins** (e.g., mannose-binding protein, MIP-133) — none is a germline cause of disease.

Summary

Acanthamoeba keratitis is a sight-threatening, painful infection of the cornea caused by free-living *Acanthamoeba* protozoa, with genotype **T4** accounting for roughly two-thirds of clinical and environmental isolates. Because the etiology is infectious rather than Mendelian, there is no OMIM gene entry, no causal variant, and no inheritance pattern; the identifiers that anchor this entry are **MONDO:0005629**, MeSH "Acanthamoeba Keratitis," and **ICD-10 B60.12 / ICD-11 1F56**. Host immune-gene polymorphisms (CXCL8/IL-8, TLR4, Th17-pathway genes) modify disease *severity* but not susceptibility.

AK is overwhelmingly a disease of **contact-lens wearers** (≈85–90% of cases in developed countries), acquired through water exposure, poor lens hygiene, and practices such as rinsing/storing lenses in tap water, "topping-off" solution, and swimming or showering in lenses. The pathogenic cascade runs from parasite adhesion to the corneal epithelium via a **mannose-binding protein (MBP)**, through secretion of a **mannose-induced 133-kDa protease (MIP-133)** and other proteases that drive corneal-cell apoptosis and stromal matrix

degradation, to intense innate/adaptive inflammation (the hallmark **ring infiltrate** and **radial keratoneuritis**), and the formation of **drug-resistant double-walled cysts** that make the disease chronic and difficult to cure.

Clinically, AK presents with **severe ocular pain disproportionate to clinical signs**, photophobia, and redness, and is frequently misdiagnosed early as **herpes simplex keratitis** (≈20% of cases), causing diagnostic delay of 3–4 weeks. Diagnosis relies on **in vivo confocal microscopy (IVCM)** and **18S rRNA PCR** (both far more sensitive than culture). Treatment is **prolonged topical cysticidal biguanide therapy** (PHMB or chlorhexidine), increasingly as higher-concentration **PHMB 0.08% monotherapy** (now the first specifically approved AK drug, marketed as Akantior), often with an aromatic diamidine; refractory/advanced cases require therapeutic keratoplasty (needed in ≈30–47% in various cohorts). Prevention is entirely behavioral. AK is rarely fatal (confined to the cornea) but is a major cause of unilateral corneal blindness, and its global incidence is rising with increasing contact-lens use.

Key Findings

Finding 1 — AK is an infectious, non-genetic corneal disease; contact-lens wear dominates risk

Acanthamoeba keratitis has **no human causal gene**. The etiologic agent is the free-living amoeba *Acanthamoeba* (genotype **T4 predominant**, ~66% of environmental and clinical isolates; PMID: 34453084: *"the most predominant genotype T4 found in 87 isolates (66.4%)"*). The dominant risk factor is contact-lens wear. In the **UK 2015 prospective national survey**, 124 AK cases gave an overall incidence of **2.35 per million**, with contact-lens wearers accounting for **108/124 (87%)** and an incidence among lens wearers of **26.94 per million** (PMID: 38097200). Risk varied sharply by lens type: planned-replacement soft lens wearers had an incidence of **50.65 per million — 7-fold higher** than daily-disposable users (7.24 per million).

A large **Netherlands case-control study** (220 cases / 1020 controls) quantified acquisition risk factors (PMID: 42393849: *"Identified risk factors were previous eye injury (OR: 16.0, 95%CI: 3.4-76.7), occupational/recreational surface water contact (OR: 9.1, 95%CI: 3.5-23.2), cleaning/rinsing CLs with tap water (OR: 3.3, 95%CI: 1.7-6.4), cigarette smoke exposure (OR: 1.9, 95%CI: 1.1-3.5) and napping in CLs (OR: 1.9, 95%CI: 1.0-3.6)."*). In pediatric **orthokeratology** wearers, inadequate hand hygiene (OR 26.46, 3.29–212.79) and "topping-off" solution (OR 17.40, 2.03–149.34) were the strongest modifiable risks (PMID: 42442142).

Risk factor	Odds ratio (95% CI)	Source
Previous eye injury	16.0 (3.4–76.7)	Netherlands [PMID: 42393849]
Surface-water contact	9.1 (3.5–23.2)	Netherlands [PMID: 42393849]
Rinsing lenses with tap water	3.3 (1.7–6.4)	Netherlands [PMID: 42393849]
Cigarette smoke exposure	1.9 (1.1–3.5)	Netherlands [PMID: 42393849]
Napping in lenses	1.9 (1.0–3.6)	Netherlands [PMID: 42393849]
Inadequate hand hygiene (ortho-k)	26.46 (3.29–212.79)	Ortho-k [PMID: 42442142]
Topping-off solution (ortho-k)	17.40 (2.03–149.34)	Ortho-k [PMID: 42442142]

Finding 2 — Pathogenesis causal chain: MBP adhesion → MIP-133 protease → cytopathic effect, apoptosis, matrix degradation

The first critical step is **adhesion** of trophozoites to the corneal epithelium via the amoebic **mannose-binding protein (MBP)**. Adhesion triggers secretion of a **contact-dependent metalloproteinase** plus **contact-independent serine proteinases** that kill host cells and degrade the basement membrane and stromal matrix (PMID: 20427010: *"Acanthamoebae express a major virulence protein, the mannose-binding protein (MBP), which mediates the adhesion of amoebae to the surface of the cornea... the amoebae produce a contact-dependent metalloproteinase and several contact-independent serine proteinases."*).

A mannose-induced **133-kDa protease (MIP-133)** facilitates corneal invasion and **induces apoptosis** via the **cytosolic phospholipase A2α (cPLA2α)** pathway with caspase-3 activation (PMID: 23132804: *"stimulates trophozoites to secrete a mannose-induced 133 kDa protease (MIP-133), which facilitates corneal invasion and induces apoptosis"*). A purified **33-kDa serine proteinase** shows cytopathic effects on HCE cells, blocked by PMSF, correlating with strain virulence (PMID: 17170574). Notably, **cysts induce greater cytotoxicity and pro-inflammatory cytokine release than trophozoites** (PMID: 31284343: *"A. castellanii induced the expression of IL-1α, IL-6, IL-8, and CXCL1 in HCECs"*).

Finding 3 — Diagnosis: IVCM and 18S rRNA PCR outperform culture; frequent misdiagnosis as herpes

In a 42-patient study, sensitivities were **nested PCR 96.3%, IVCM 77.78%, one-step PCR 62.96%, culture 37.04%**, with specificity and PPV of 100% for all tests (PMID: 41588033: *"the sensitivity of IVCM, culture, PCR, and nested PCR was 77.78%, 37.04%, 62.96%, and 96.3%, respectively. The specificity and positive predictive values were 100% for all the tests."*). IVCM reveals **double-walled hyperreflective cysts**; PCR/sequencing targets the **18S rRNA gene**. Diagnostic delay is a defining problem: **herpes simplex keratitis was the initial misdiagnosis in 20.2% (25/124)** of UK cases (PMID: 38097200: *"Herpes keratitis was initially misdiagnosed in 25 of 124 cases (20.2%)."*). Multiplex PCR can yield **false positives** in patients lacking classic risk factors (PMID: 42949848).

Diagnostic test	Sensitivity	Specificity / PPV
Nested PCR (18S rRNA)	96.3%	100%
In vivo confocal microscopy	77.8%	100%
One-step PCR	63.0%	100%
Culture (non-nutrient agar)	37.0%	100%

Finding 4 — Treatment: prolonged biguanide ± diamidine; PHMB 0.08% monotherapy and protocol delivery improve cure

The mainstay is **prolonged combination therapy with topical biguanides (PHMB or chlorhexidine) and aromatic diamidines**, with growing evidence for **PHMB 0.08% monotherapy** (PMID: 42262140: *"prolonged combination therapy with topical biguanides and aromatic diamidines remains the mainstay of treatment. However, recent evidence supports the use of topical biguanide monotherapy (in higher concentrations) in AK with comparable outcomes to combination treatment."*). An indirect comparison found 12-month clinical resolution of **84.8% for PHMB 0.08%** versus 43.6% for any anti-amoebic, 55.0% for PHMB 0.02%+diamidine, and 40.0% for chlorhexidine ± diamidine (PMID: 40659372). **Protocol-based delivery** gave a **1.59-fold higher medical cure rate** vs individualized treatment (95% CI 1.40–1.80, p<0.001) (PMID: 40204130).

Regimen	12-month clinical resolution rate
PHMB 0.08% monotherapy	84.8%
PHMB 0.02% + diamidine	55.0%
Any anti-amoebic treatment	43.6%
Chlorhexidine 0.02% ± diamidine	40.0%

Finding 5 — Clinical phenotype: severe pain out of proportion to signs, ring infiltrate, radial keratoneuritis

Hallmark features are **severe ocular pain disproportionate to clinical findings**, photophobia, and treatment-resistant cysts (PMID: 42885963: *"characterized by severe pain disproportionate to clinical findings and the presence of cyst forms that confer significant treatment resistance."*). Characteristic signs include a **ring-shaped stromal infiltrate** and **radial keratoneuritis**. **Hypopyon is uncommon in pure AK** and, when present, often signals fungal coinfection with worse prognosis (PMID: 42685305: *"AK with hypopyon can be a representative sign of fungal coinfection, which has a trend of a worse visual prognosis."*). Most cases are **unilateral**; bilateral disease is rare.

Finding 6 — Prognosis: ~30–47% require keratoplasty; older age, longer symptom duration, ring infiltrate predict poor outcome

In a 51-eye cohort, **15/51 (29.4%) required therapeutic keratoplasty**; predictors were older age (OR 1.09/yr) and longer symptom duration (OR 1.03/day) (PMID: 42301488: *"Fifteen eyes (29.4%) required TK... older age at diagnosis (OR 1.09; 95% CI 1.03-1.16; p = 0.005) and longer symptom duration (OR 1.03; 95% CI 1.01-1.05; p = 0.011) were associated with TK."*). In the Netherlands cohort, **treatment failure occurred in 85/179 (47%)** and globe loss in 5% (PMID: 42393849). The Danish genotyped cohort confirmed poor-outcome predictors (pre-diagnosis corticosteroids, moderate-severe pain, male sex, older age, ring infiltration, infiltrate >3 mm) while **genotype did not predict outcome** (PMID: 40387241).

Finding 7 — Host immune-gene polymorphisms modify severity

In 105 AK cases (38% with severe inflammatory complications), **CXCL8 (IL-8) SNPs were protective** (rs1126647 OR=0.3, p=0.005), **two TLR4 SNPs increased risk** (both OR=6.9, p=0.01), and **Th17 SNPs** (IL23R, IL1B, IL22) were associated with complications (PMID: 33755043). In

mouse models, **IL-17A is protective** (PMID: 25505284: *"corneal Acanthamoeba infection induces IL-17A expression and that IL-17A is critical for host protection against severe AK pathology"*). The C-type lectin receptor **CLEC4E (Mincle)** mediates protective macrophage NF-κB activation (PMID: 42929688).

Finding 8 — Model organisms and vaccine proof-of-concept

The **Chinese hamster** model develops AK via *A. castellanii*-laden lenses; **oral MBP immunization ameliorates disease** with elevated anti-MBP tear IgA (PMID: 16982837: *"oral immunization with recombinant MBP ameliorates AK in a hamster animal model and that this protection is associated with the presence of elevated levels of anti-MBP immunoglobulin A in the tear fluid..."*). Oral **MIP-133/aPA–cholera toxin** immunization confers additive protection (PMID: 18055809). The **mouse corneal model** dissects IL-17A, DECTIN-1, and CLEC4E immunity.

Finding 9 — Etiologic agent: *Acanthamoeba* spp. with trophozoite and resistant cyst stages

Acanthamoeba is ubiquitous in tap water, soil, and pools (PMID: 39764873: *"Acanthamoeba castellanii is a widespread unicellular eukaryote found in diverse environments, including tap water, soil, and swimming pools."*). The **dormant, double-walled cyst** is highly drug-resistant; encystation involves **global metabolic downregulation** (PMID: 40738523: *"a global metabolic downregulation during encystation, which is consistent with a state of dormancy. Components of the cyst wall such as cellobiose and lactose accumulated in the final phase."*) and **autophagy** (PMID: 25896709). *Acanthamoeba* can act as a **"Trojan horse"** for intracellular bacteria (PMID: 23444840).

Finding 10 — Temporal course: epithelial-stage onset progressing to stromal ring infiltrate

AK begins as **superficial epitheliopathy** progressing to the stroma (PMID: 35402748). Early signs can be subtle/atypical, promoting herpes misdiagnosis (PMID: 34348354). In the mouse model, temporal transcriptomics showed **peak inflammation at 3 days post-infection**, early innate then adaptive immunity, with DECTIN-1/NF-κB throughout (PMID: 42872947: *"AK progression was characterized by dynamic changes, with peak inflammatory responses at 3 dpi. Immune infiltration analysis revealed early innate immune activation followed by adaptive immune involvement at later stages."*).

Finding 11 — Demographics: bimodal age, male predominance, tap-water lens exposure

The CDC multistate study (88 cases/151 controls) found cases more likely aged **<25 (mOR 2.7) or >53 (mOR 2.5)** and **male (mOR 2.6)** (PMID: 28099282: *"Case patients were more likely to be aged <25 years (unadjusted mOR 2.7, 95% confidence interval 1.3-5.5) or aged >53 years (mOR 2.5, 1.1-5.7), and more likely to be men (mOR 2.6, 1.4-4.8)."*). The Chicago outbreak cohort reported mean age 28±15, 52.5% men, 95% lens wearers (PMID: 16876498).

Finding 12 — *Acanthamoeba* also causes GAE and cross-species disease

Beyond keratitis, *Acanthamoeba* causes **granulomatous amebic encephalitis (GAE)**, a rare, nearly universally fatal CNS infection in immunocompromised hosts (PMID: 42158383: *"Granulomatous amebic encephalitis (GAE) is a rare but frequently fatal infection of the central nervous system..."*). AK itself is confined to the cornea and rarely fatal. Naturally occurring AK has been documented in **wild birds** (PMID: 28807756: *"Acanthamoeba infection was detected in wild bird cornea samples with/without keratitis for the first time in the world."*).

Finding 13 — Rising global incidence; emerging adjunctive therapies

AK has *"evolved from an extraordinarily rare disease to a significant public health concern, with increasing global incidence"* (PMID: 41111636). **PHMB 0.08% (Akantior)** is the first approved AK drug (90.9% resolution in a Polish cohort, PMID: 42961683). Adjuncts include voriconazole, miltefosine, and **PACK-CXL** (PMID: 39748383: *"The same-session combination of riboflavin/UV-A and rose bengal/green light PACK-CXL effectively treated a patient with confirmed AK that was resistant to conventional medical therapy"*; PMID: 42348670). Refractory T4 cases can still progress to **phthisis bulbi** (PMID: 41526733).

Section-by-Section Report

1. Disease Information

AK is a sight-threatening, painful corneal infection caused by *Acanthamoeba*. Key identifiers: **MONDO:0005629**; MeSH "Keratitis, Acanthamoeba"; **ICD-10 B60.12**; **ICD-11 1F56**. No OMIM entry (non-genetic). Synonyms: *Acanthamoeba caused keratitis, keratitis, Acanthamoeba,* amoebic keratitis. Information is derived from **aggregated disease-level resources** (surveillance surveys, case-control studies, case series), not single-patient EHR.

2. Etiology

Causal factor: infectious — *Acanthamoeba* spp. (T4 predominant); **no genetic cause**. **Environmental risks**: contact-lens wear (dominant), water exposure of lenses, poor hygiene, topping-off, trauma, hard domestic water, smoke exposure (Findings 1, 11). **Host genetic *modifiers*** (not causes): protective CXCL8/IL-8, risk-increasing TLR4/Th17 (Finding 7). **Protective factors**: daily-disposable lenses, strict hygiene; mucosal IgA and milk proteinase inhibitors reduce cytopathic effect in vitro (PMID: 18326724). **Gene–environment interaction**: host immune background tunes the *severity* of inflammation to an environmentally acquired infection.

3. Phenotypes

Phenotype	Type	Frequency/severity	HPO suggestion
Severe pain disproportionate to signs	Symptom	Very frequent; severe	eye pain
Photophobia	Symptom	Frequent	HP:0000613
Ring stromal infiltrate	Clinical sign	Common; severity marker	HP:0200136 (keratitis)
Radial keratoneuritis	Clinical sign	Characteristic/early	—
Reduced visual acuity	Sign	Frequent	HP:0000572 / HP:0000505
Conjunctival hyperemia (redness)	Sign	Frequent	HP:0000532
Epitheliopathy / pseudodendrites	Sign	Early stage	—

Onset is **adult** (contact-lens wearers, 20–40), bimodal peaks <25 and >53. Progression is **progressive if untreated**. QoL impact is substantial: chronic pain, months-long therapy, vision loss (PMID: 33151784).

4. Genetic/Molecular Information

Not applicable as a causal mechanism — no human causal gene, no pathogenic variants, no chromosomal abnormality, no inheritance. The only relevant human genetics are **severity modifiers** (CXCL8/IL-8 protective; TLR4, IL23R, IL1B, IL22 risk — Finding 7). The pathogen's molecular determinants (MBP, MIP-133, serine proteases, cyst-wall genes) are the functional molecular information (Findings 2, 9).

5. Environmental Information

Environmental agents: waterborne *Acanthamoeba* (tap water, pools, soil, biofilms). **Lifestyle**: contact-lens wear/hygiene; smoking is associated (PMID: 42393849). **Infectious agent**: *Acanthamoeba* spp. (NCBI taxid 5754), T4 predominant; can co-infect with bacteria/fungi.

6. Mechanism / Pathophysiology — Causal Chain

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1. Environmental exposure: Acanthamoeba trophozoites/cysts contact the ocular surface
   (contaminated contact lens, water, or trauma)
        | leads to
2. ADHESION: amoebic mannose-binding protein (MBP) binds mannosylated corneal
   epithelial glycoproteins (facilitated by lens-related epithelial microtrauma)
        | triggers
3. PROTEASE SECRETION: MIP-133 (133 kDa), a contact-dependent metalloproteinase,
   and contact-independent serine proteinases (e.g., 33 kDa)
        | results in (branch)
   ├─4a. HOST-CELL DEATH: MIP-133 → cPLA2α → arachidonic acid → caspase-3 →
   |      apoptosis of corneal epithelial cells; direct cytopathic effect
   └─4b. MATRIX DEGRADATION: proteases digest basement membrane, laminin/fibronectin,
          stromal collagen → amoebic invasion into deep stroma
        | provokes
5. INFLAMMATION: damaged epithelium + cysts release IL-1α/β, IL-6, IL-8/CXCL8, CXCL1;
   PRRs (DECTIN-1, CLEC4E/Mincle) → SYK/NF-κB (peak ~3 dpi in model);
   neutrophil then CD4+ Th17 (IL-17A protective) infiltration
        | manifests as
6. CLINICAL DISEASE: radial keratoneuritis (perineural invasion) + ring stromal
   infiltrate, severe pain disproportionate to signs, photophobia, epithelial defects
        | sustained by
7. CHRONICITY/RESISTANCE: trophozoites encyst (autophagy-dependent) into double-walled,
   dormant, drug-resistant cysts → relapse, prolonged therapy, stromal necrosis
        | can progress to
8. OUTCOME: scarring, melting/perforation → therapeutic keratoplasty,
   or phthisis bulbi/vision loss in refractory cases
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Molecular pathways: cPLA2α–arachidonic acid–caspase-3 apoptosis; SYK–NF-κB inflammation downstream of DECTIN-1/CLEC4E; HIF-1α amplification. **Cellular processes (GO)**: apoptotic process (GO:0006915), inflammatory response (GO:0006954), autophagy (GO:0006914, parasite), proteolysis (GO:0006508), extracellular matrix disassembly (GO:0022617). **Cell types (CL)**: corneal epithelial cell (CL:0000575), keratocyte (CL:0002363), macrophage (CL:0000235), neutrophil (CL:0000775), Th17 cell (CL:0000899). Upstream events are parasite adhesion/proteolysis; downstream events are host inflammation and tissue destruction.

7. Anatomical Structures Affected

Primary organ: cornea (UBERON:0000964), eye (UBERON:0000970). **Tissues**: corneal epithelium (UBERON:0001772), corneal stroma (UBERON:0001777), corneal nerves; occasionally sclera. **Cells**: corneal epithelial cells (CL:0000575), keratocytes (CL:0002363).

Subcellular (GO CC): plasma membrane, cytosol, extracellular region. **Localization**: central/paracentral cornea; **usually unilateral**. Secondary: anterior chamber inflammation, hypopyon (often signaling coinfection), globe loss in extreme cases.

8. Temporal Development

Onset: adult, acute-to-subacute. **Stages**: I epithelial/limbitis → II–III stromal (ring infiltrate) → late ring/perforation; most present at stage ≥II. **Duration**: often months (Icelandic 5–24 months; Polish mean 188 days to resolution). **Critical window**: early diagnosis and cysticidal therapy before stromal/ring involvement drives good outcomes (Findings 6, 10).

9. Inheritance and Population

Epidemiology: 1–2.35 per million general population; ~17–27 per million among lens wearers (UK) with marked geographic variation (higher in hard-water regions; PMID: 11973250). **Inheritance: none (infectious)** — penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, carrier frequency are **not applicable**. **Demographics**: predominantly young adult lens wearers (20–40), bimodal peaks <25 and >53, slight male predominance; rising global incidence.

10. Diagnostics

Clinical/lab: corneal scraping with non-nutrient agar culture (low sensitivity ~37%); **18S rRNA / nested PCR** (up to 96%); **IVCM** (double-walled cysts, ~78%); AS-OCT; histopathology (PAS/calcofluor white). **Parasite genotyping** by 18S rRNA sequencing (T4 predominant). **Clinical criteria**: compatible history + disproportionate pain + characteristic signs + micro/IVCM confirmation. **Differential diagnosis**: herpes simplex keratitis (commonest misdiagnosis), fungal, bacterial. Multiplex PCR false positives require clinical correlation (PMID: 42949848). **Screening**: none; prevention is behavioral.

11. Outcome/Prognosis

Rarely fatal; high morbidity (chronic pain, scarring, vision loss). Keratoplasty in **~29–47%**; globe loss ~5%. **Prognostic factors**: older age, longer symptom duration/diagnostic delay, ring infiltrate, infiltrate >3 mm, pre-diagnosis corticosteroids, male sex; genotype not prognostic. Historical pre-propamidine medical cure was only 19.6% vs much improved modern outcomes (PMID: 40323557).

12. Treatment

Pharmacotherapy (NCIT-type intervention terms): topical **biguanides** — PHMB/polihexanide 0.02%–0.08%, **chlorhexidine** 0.02%–0.2%; **aromatic diamidines** — propamidine isethionate 0.1%, hexamidine. **PHMB 0.08% (Akantior)** is the first approved AK drug, effective as monotherapy. **Adjuncts**: voriconazole, **miltefosine**, povidone-iodine, antimicrobial peptides, **PACK-CXL** for refractory disease. Corticosteroids used cautiously. **Surgical**: therapeutic/penetrating keratoplasty for refractory/perforating disease. **Strategy**: protocol-based dual-therapy dosing improves cure 1.59-fold (PMID: 40204130). Pharmacogenomics: not established.

13. Prevention

Entirely **behavioral/primary**: contact-lens hygiene, **avoid water (tap/pool/shower/swim) contact with lenses**, no topping-off, hand hygiene, timely case replacement, prefer daily-disposables, prompt evaluation of ocular discomfort (PMID: 42442142, PMID: 28099282). **Secondary**: early recognition to avoid herpes misdiagnosis/delay. **Public health**: patient/practitioner education; water-management awareness. No human vaccine (animal MBP/MIP-133 immunization is protective — Finding 8).

14. Other Species / Natural Disease

Pathogen taxonomy: *Acanthamoeba* spp. (NCBI taxid 5754). **Natural disease**: AK in **wild birds** (Eurasian sparrowhawk, *Accipiter nisus*; T4/T5) (PMID: 28807756); *Acanthamoeba* also causes GAE/disseminated disease across animals. **No Mendelian animal homolog** (OMIA not applicable). **Transmission**: environmentally acquired; **not zoonotic**.

15. Model Organisms

Chinese hamster (*Cricetulus griseus*) — contact-lens corneal infection; oral MBP/MIP-133/aPA immunization studies (PMID: 16982837, PMID: 18055809). **Mouse** — corneal intrastromal-injection model; IL-17A (PMID: 25505284), DECTIN-1 (PMID: 42872947), CLEC4E (PMID: 42929688). **In vitro**: HCEC, MDCK monolayers, ex vivo cornea. **Recapitulation**: reproduce corneal infection, protease cytopathology, inflammation, and (hamster) protective mucosal immunity. **Limitations**: corneal anatomy/immunity differ from humans; models do not fully reproduce chronic cyst-driven relapse.

Mechanistic Model / Interpretation

AK is best understood as a **two-phase infectious process**: an initial *parasite-driven cytopathic phase* (adhesion → proteolysis → apoptosis and matrix destruction) followed by a *host inflammatory phase* (PRR-driven NF-κB inflammation, neutrophil and Th17 responses) that produces the ring infiltrate and much of the tissue damage. The **cyst stage** is the linchpin of clinical difficulty: its dormant metabolism and resistant double wall explain treatment failure, prolonged therapy, and relapse. Host genetics tune the *intensity* of phase two but do not determine *whether* infection occurs — that is set by environmental exposure and lens hygiene.

The clinical corollaries all flow from this model: (1) **early diagnosis matters** because intervention during the epithelial phase yields the best outcomes; (2) **cysticidal biguanides** are first-line because they act on the resistant cyst; (3) **corticosteroids before diagnosis worsen outcomes** by suppressing protective immunity while the parasite persists; and (4) **prevention is behavioral** because exposure, not the genome, is modifiable.

Evidence Base

PMID	Role	Key contribution
38097200	Supports F1, F3	UK national incidence; lens risk gradient; 20.2% herpes misdiagnosis
42393849	Supports F1, F6	Acquisition risk ORs; 47% treatment failure; prognostic factors
34453084	Supports F1, F9	T4 genotype predominance (66.4%)
20427010	Supports F2	MBP adhesion and protease cascade
23132804	Supports F2	MIP-133, cPLA2α apoptosis
31284343	Supports F2	Cytokine induction; cysts more cytopathic
41588033	Supports F3	Head-to-head diagnostic sensitivities
40659372	Supports F4	Comparative clinical resolution rates
42262140	Supports F4, F13	Mainstay + monotherapy; emerging therapies
40204130	Supports F4	Protocol delivery improves cure 1.59×
42885963	Supports F5	Disproportionate pain; cyst resistance
42685305	Supports F5	Hypopyon signals fungal coinfection
42301488	Supports F6	Keratoplasty rate; TK predictors
40387241	Supports F6	Clinical predictors; genotype not prognostic
33755043	Supports F7	Host immune-gene SNP modifiers
25505284	Supports F7, F15	IL-17A protective (mouse)
42929688	Supports F7, F15	CLEC4E/HIF-1α macrophage defense
16982837	Supports F8	Hamster model; MBP oral immunization
18055809	Supports F8	MIP-133/aPA immunization
39764873	Supports F9	Environmental ubiquity
40738523	Supports F9	Encystation metabolic dormancy

PMID	Role	Key contribution
35402748	Supports F10	Epithelial-stage disease
42872947	Supports F10	DECTIN-1 temporal immunity (3 dpi peak)
28099282	Supports F11	CDC demographics & modifiable risk
16876498	Supports F11	Chicago outbreak demographics
42158383	Supports F12	GAE (fatal CNS form)
28807756	Supports F12	AK in wild birds
41111636	Supports F13	Rising global incidence
39748383	Supports F13	PACK-CXL for refractory AK
42348670	Supports F13	Adjunctive therapies

Limitations and Knowledge Gaps

1. **Observational evidence base.** Most epidemiology/prognosis data come from case series and case-control studies, susceptible to referral and recall bias; randomized treatment trials remain few (ODAK is a notable exception).
2. **No universally adopted staging or diagnostic guidelines**, contributing to diagnostic delay and heterogeneous outcome reporting.
3. **Cyst biology** — molecular triggers and regulatory networks of encystation/excystation are incompletely defined, limiting rational cysticidal drug design.
4. **Host-genetics modifiers** rest largely on a single ~105-patient study and animal models; need replication in larger, diverse cohorts.
5. **No human vaccine** despite promising animal proof-of-concept; translation barriers unaddressed.
6. **Molecular false positives** (multiplex PCR) and the absence of validated viability assays complicate diagnosis and test-of-cure.

Proposed Follow-up Experiments / Actions

1. **Prospective multicenter registry** with standardized staging, genotyping, and outcome definitions to firm up prognostic models and treatment comparisons.
2. **Randomized trials** comparing PHMB 0.08% monotherapy vs combination regimens and quantifying adjunct value (miltefosine, PACK-CXL) in refractory disease.
3. **Replication of host immune-gene associations** (CXCL8, TLR4, Th17) with functional validation.
4. **Cyst-targeted drug discovery** leveraging encystation metabolomics/autophagy dependence; develop **viability PCR / cyst-viability assays** for test-of-cure.
5. **Diagnostic standardization**: validate AI-assisted IVCM and metagenomic sequencing; define criteria to avoid multiplex-PCR false positives.
6. **Translational vaccine work**: advance MBP/MIP-133 mucosal immunization toward human-relevant platforms; evaluate prophylaxis in high-risk lens wearers.
7. **Public-health intervention trials** on lens-hygiene education and water-avoidance, especially in hard-water regions and pediatric ortho-k populations.
